THE MECHANICS OF SURVIVAL

Cardiac Hydraulics, Acid-Base Alchemy, & Artificial Bellows

DIRECTIVES FOR ADVANCED PHYSIOLOGY INITIATES

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The Hydraulic Integrity Check

Quantifying Pump Stress vs. Engine Failure

Biomarker Analysis: Pressure Overload (BNP) & Myocardial Damage (Troponin).

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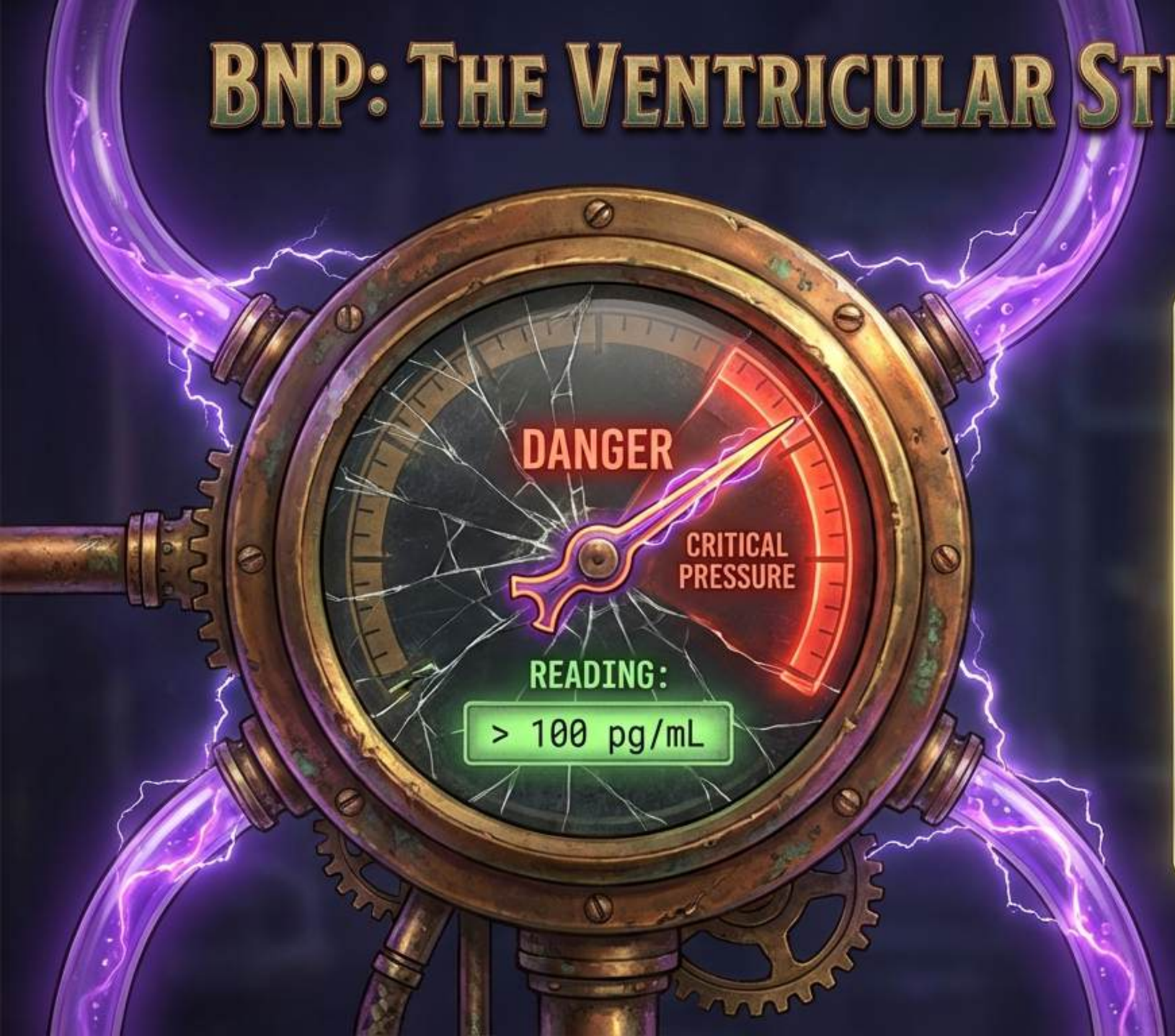
BNP: The Ventricular Stretch Gauge



SECRETION TRIGGER:
Excessive Ventricular Stretching.

INDICATOR:
Congestive Heart Failure (CHF).

NORMAL RANGE:
< 100 pg/mL (Safe Pressure).

CRITICAL READOUT:
> 100 pg/mL (Hydraulic Overload).

PROTOCOL:
Unload the System. Administer Loop Diuretics (Furosemide).

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TROPONIN: THE ISCHEMIC THRESHOLD

MECHANISM: Myocardial Necrosis (Engine Block Damage).
NORMAL: Troponin I < 0.035 ng/mL.
CRITICAL: Any Elevation = Myocardial Infarction.
DIFFERENTIAL DIAGNOSIS:
- BNP = STRETCH (Chronic Load)
- TROPONIN = DEATH (Acute Ischemia)

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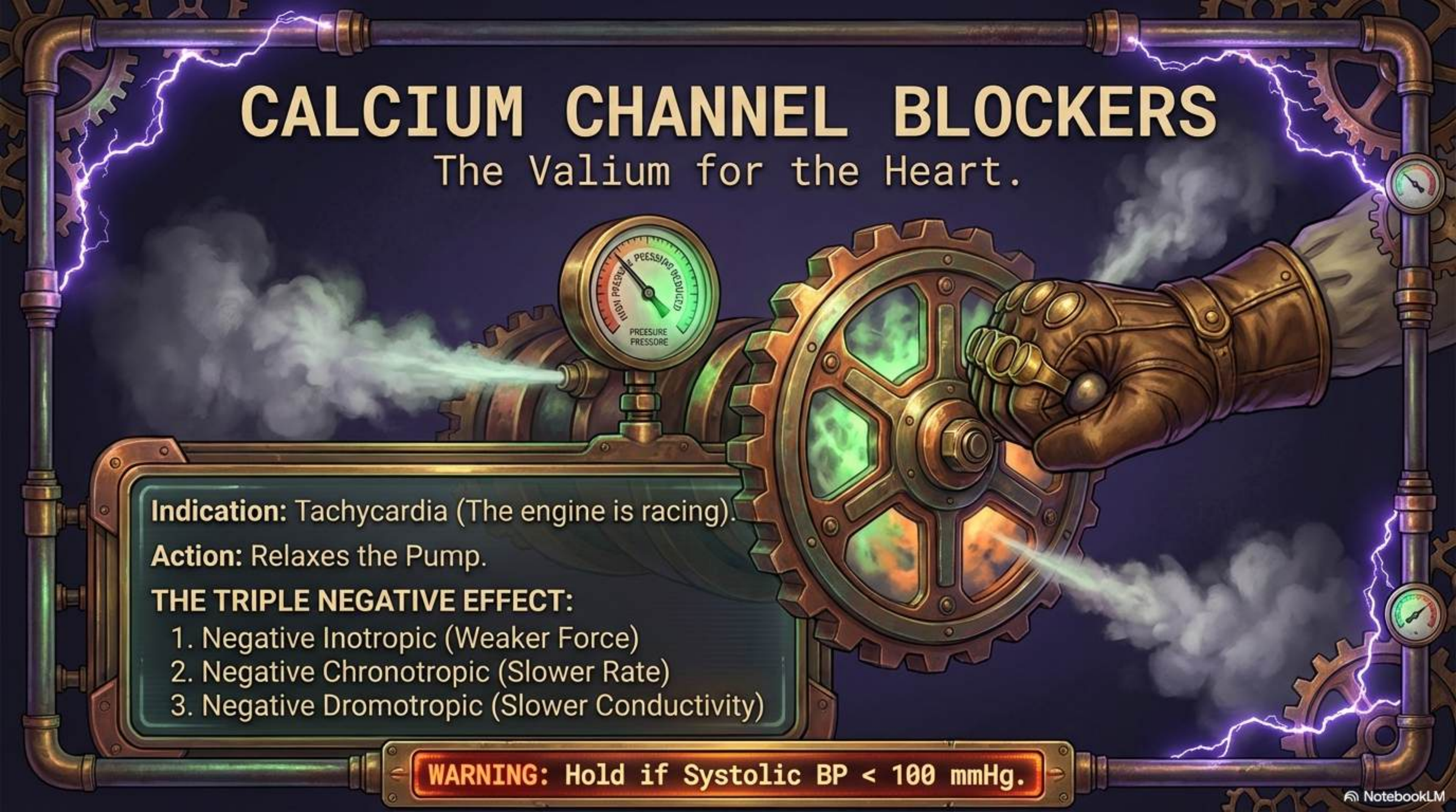
CALCIUM CHANNEL BLOCKERS

The Valium for the Heart.



Indication: Tachycardia (The engine is racing).

Action: Relaxes the Pump.

THE TRIPLE NEGATIVE EFFECT:

1. Negative Inotropic (Weaker Force)
2. Negative Chronotropic (Slower Rate)
3. Negative Dromotropic (Slower Conductivity)

WARNING: Hold if Systolic BP < 100 mmHg.

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ELECTRICAL FAILURE: LETHAL RHYTHMS

V-FIB: CHAOTIC / NO OUTPUT.

PROTOCOL: D-Fib (Defibrillation).

ASYSTOLE: SPARK FAILURE.

PROTOCOL: Epinephrine + Atropine.

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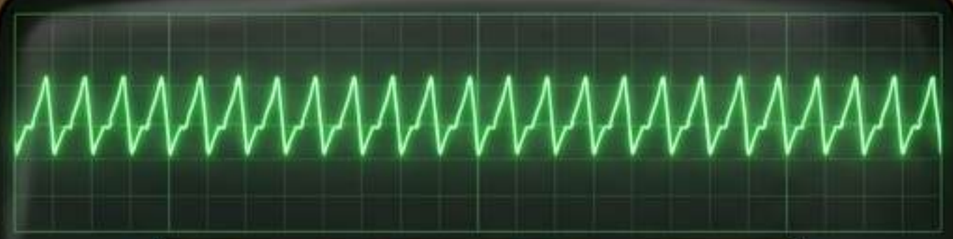
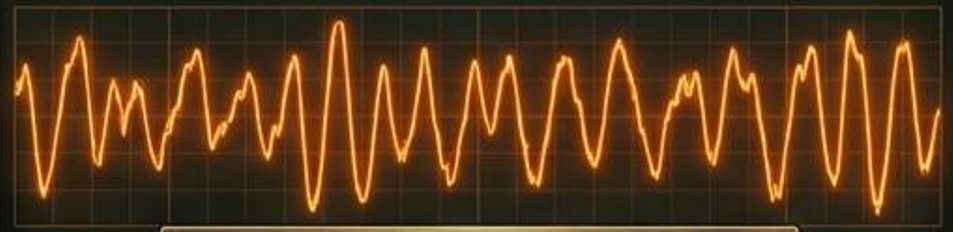
Electrical Failure: Ventricular & Atrial



ATRIAL FLUTTER (Sawtooth)



V-TACH (Patterned Chaos)

PVCs / V-Tach Protocol:
Amiodarone or Lidocaine.

Atrial Arrhythmia Protocol (ABCD):

A: Adenocard (Fast Push)
B: Beta Blockers
C: Calcium Channel Blockers
D: Digitalis

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Rye

ACID-BASE ALCHEMY

pH (MASTER GAUGE):

7.35 - 7.45

PaCO2 (LUNG EXHAUST):

35 - 45 mmHg

HCO3 (KIDNEY FILTER):

22 - 26 mEq/L

Roboto Mono

The machine operates on a precise fuel mixture.
Mathematical deduction is required.

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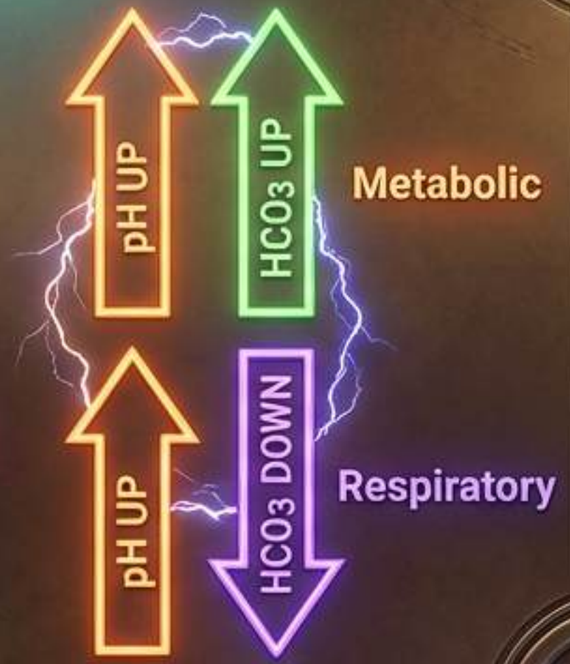
THE RULE OF THE B's

If the pH and the Bicarb are BOTH in the same direction, then it is MetaBOLIC.

pH DOWN + HCO_3 DOWN = Metabolic Acidosis
pH UP + HCO_3 UP = Metabolic Alkalosis
Opposite Directions = Respiratory



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Rye

SYMPTOM PREDICTION: THE pH VELOCITY

As the pH goes, so goes my patient.

ALKALOSIS (pH UP)

HCO_3 UP

$PaCO_2$ DOWN

- Irritability
- Hyper-reflexia (+3/+4)
- Tachypnea
- Tachycardia
- Seizures.

ACIDOSIS (pH DOWN)

pH UP

HCO_3 DOWN

- Lethargy
- Bradycardia
- Hypo-reflexia (+1)
- Paralytic Ileus
- Coma.

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Rye

THE INVERSE POTASSIUM LAW

As the pH goes, so goes my patient, EXCEPT for Potassium.

pH DOWN (Acidosis) = Potassium UP (Hyperkalemia)
Risk: Cardiac Arrest.

pH UP (Alkalosis) = Potassium DOWN (Hypokalemia)
Risk: Muscle Cramps / Arrhythmias.

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RESPIRATORY DIAGNOSTICS: THE BELLOWS TEST

Step 1: Is it Lung? → YES → Respiratory

Step 2: Over-ventilating or Under-ventilating?

Over-ventilating (Fast)
= Blowing off CO_2
= ALKALOSIS

Under-ventilating (Slow)
= Retaining CO_2
= ACIDOSIS

Examples: Overdose/COPD = Under-ventilating = Acidosis

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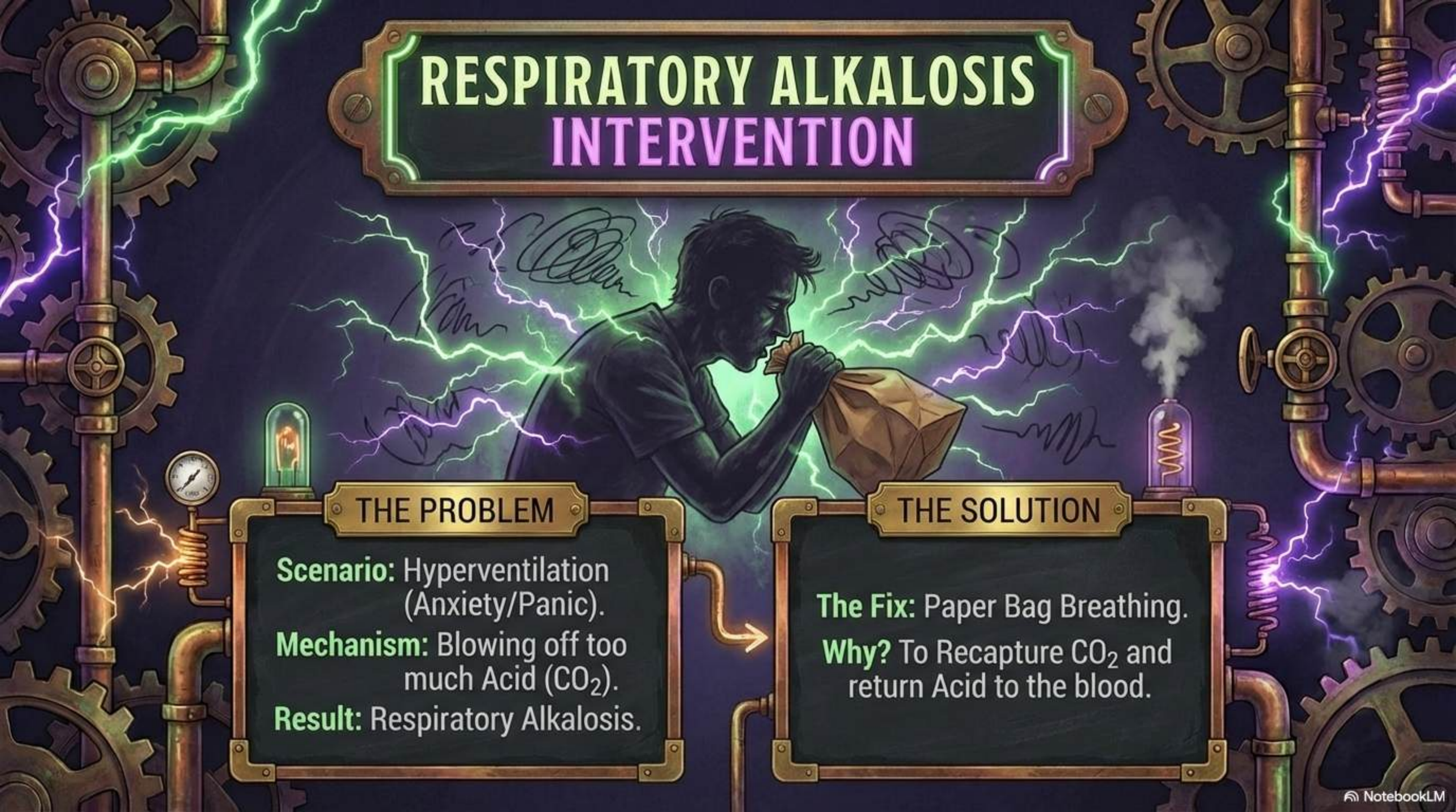
RESPIRATORY ALKALOSIS
INTERVENTION



THE PROBLEM

Scenario: Hyperventilation (Anxiety/Panic).

Mechanism: Blowing off too much Acid (CO_2).

Result: Respiratory Alkalosis.

THE SOLUTION

The Fix: Paper Bag Breathing.

Why? To Recapture CO_2 and return Acid to the blood.

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THE COMPENSATORY GASP: MAC KUSSMAUL

Deep, Labored Breathing Pattern = KUSSMAUL RESPIRATIONS.

The Law: Kussmaul is **ONLY** seen in Metabolic Acidosis (MAC).

Metabolic Diagnostics:

1. Vomiting/Suction = Losing Acid = Metabolic Alkalosis.
2. Diarrhea = Losing Base = Metabolic Acidosis.
3. **Default Rule:** If it's not Lung, and not Vomiting, pick Metabolic Acidosis.

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ARTIFICIAL BELLOWS: ALARM PROTOCOLS

HIGH PRESSURE ALARM

Cause: Increased Resistance (Obstruction).

1. Unkink Tube.
2. Empty Water.
3. Clear Mucus (Turn/Cough).
4. Suction.

LOW PRESSURE ALARM

Cause: Decreased Resistance (Disconnection).

1. Reconnect Main Tubing.
2. Reconnect O2 Sensor.

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THE SURVIVAL CONSOLE: TRIAGE HIERARCHY

MAC KUSSMAUL

(A) LOW PRIORITY

Abnormal but expected.

Action: Do nothing. (e.g., Creatinine).

(B) MONITOR

Be concerned.

Action: Assess. (e.g., BNP, Sodium).

(C) CRITICAL

Action Required.

Hold/Assess/ Prepare/Call. (e.g., INR > 4).

(D) DEADLY

DO SOMETHING NOW.

Stay with patient. (e.g., pH 6, K+ 6).

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LEVEL C ALERTS: COAGULATION

INR > 4.0

Meaning: Hemorrhage Risk (Warfarin Overdose).

Protocol:

1. **HOLD** Warfarin.
2. **ASSESS** for Bleeding.
3. **PREPARE** Vitamin K.
4. **CALL** Physician.

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LEVEL C ALERTS: IMMUNOCOMPROMISE

SEVERE NEUTROPENIA

Values: WBC < 5,000 | ANC < 500 | CD4 < 200 (AIDS).

Protocol: NEUTROPENIC PRECAUTIONS.

- Strict Hygiene.
- No Fresh Flowers or Potted Plants.
- No Raw Fruits/Vegetables.
- No Standing Water (No Pitchers >15 mins).
- No Sick Visitors.

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LEVEL D: IMMEDIATE SYSTEM FAILURE (PART I)

POTASSIUM (K+) > 6.0

Result: Cardiac Standstill.

Action: STAY with patient. Prepare Kayexalate + D5W/Insulin. Call MD.

pH in the 6s (e.g., 6.8)

Result: Severe Acidosis (Incompatible with life).

Action: Assess vitals. Call MD. Treat underlying cause immediately.

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OXYGEN
ZERO
20
40
60
80
100
120
LEVEL D: IMMEDIATE SYSTEM FAILURE (PART II)
CO2 > 60 mmHg: Respiratory Failure.
Action: Prepare to Intubate/Ventilate.
Roboto Mono
PO2 < 60 mmHg: Severe Hypoxia.
Action: Intubate/Ventilate.
Roboto Nono
Platelets < 40,000: Spontaneous Hemorrhage.
Action: Bleeding Precautions. Transfuse.
Roboto Mono
THE MASTER MECHANIST'S FINAL DIRECTIVE: TREAT THE PATIENT, NOT THE MONITOR.

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